

Rotating Spiral Brain Waves Act as a Space-and-Time Clock

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Summary: Researchers identified a new class of traveling brain waves that rotate over space and time. The study reveals that these vortex-like waves are driven by a unique, circular “merry-go-round” architectural layout of neurons in the sensory cortex.

Operating globally, these spiral waves synchronize activity across hemispheres, between sensory and motor networks, and down into deep subcortical structures—acting as a spatiotemporal clock to coordinate sensation, predict sequences, and guide voluntary physical action.

Key Facts

- **Discovery of Spiral Waves:** Scientists discovered a new category of traveling brain waves that physically rotate over space and time across the cerebral cortex.

- **“Merry-Go-Round” Wiring:** The wave’s circular motion is driven by a unique, fixed architectural layout of neurons in the somatosensory cortex whose axons point in a physical circle.
- **Cross-Network Coordination:** These vortex-like waves travel across boundaries, mirroring perfectly in both hemispheres and linking the sensory cortex to the motor cortex and deep subcortical structures.
- **Behaviorally Triggered:** A slight puff of air to a mouse’s facial whiskers instantly evoked a sequence of clockwise rotating waves, shifting in shape based on the animal’s task performance and arousal levels.
- **Spatiotemporal Clock Function:** Researchers hypothesize that these streaming waves act as a neural clock to sequence sensation followed by action, helping the brain predict sensory sequences and entrench motor skills.

Source: Washington University

Spiraling waves of neural activity appear and travel in the brain. Scientists hope to learn if these rotating waves on-the-move play a global role in sensing and interpreting internal and external stimuli, in laying down memory, and in managing motor performance.

“We discovered a new kind of brain wave that specifically rotates over space and time, relies on a circular

anatomical circuit in the sensory cortex, and impacts activity across the brain,” noted Nick Steinmetz, associate professor of neurobiology and biophysics at the University of Washington School of Medicine in Seattle. His team led the research.



Traveling brain waves form rotating spiral patterns guided by a circular anatomical arrangement of neurons, coordinating global sensory and motor networks. Credit: Neuroscience News

Details on these traveling, whirling brain waves, as well as data on their activity during certain behaviors in mice, are reported this week in *Science*.

The findings on these vortex-like waves are, as they say, head-spinning.

The scientists examined how a mouse brain's anatomical wiring coordinates the structure and propagation of the waves, which most commonly originate in the somatosensory region. This area processes sensations felt by the skin and muscles and cues about the body's position, posture and parts, as well as other stimuli.

The neurons that generate these rotating waves form a merry-go-round-like pattern in the brain's sensory cortex. Their axons, which produce electrical signals, point in a circle. This fixed architectural arrangement, almost like rail cars along a round track, coincides with the brain wave's spiral motion.

The waves were mirrored on both sides of the mouse brain and coordinated between both sensory and motor parts of the brain. The scientists observed that the spiral waves also timed with spiking detected in deeper areas of the brain associated more with low-level functions. These include the thalamus, striatum and midbrain.

Because these rotating waves travel to different brain regions, they may play a role in sharing information across parts of the brain responsible for different but interdependent functions. For example, the interplay between the sensory cortex and the motor cortex of the

brain is likely crucial to navigating one's surroundings and other voluntary physical movements.

The scientists conducted their studies using cortex-wide brain imaging and large-scale electrophysiology measurements.

Among their approaches were to see the effects of a tiny puff of air on mouse's left facial whiskers. This stimulus evoked a sequence of clockwise rotating waves of neural activity in the right sensory cortex with corresponding waves in the motor cortex.

The scientists also encouraged mice with a reward for an object-detection game that required paw and eye coordination. The scientists noticed rotating brain wave differences that varied depending on the mouse's arousal state and its success at performing the task.

The researchers have yet to determine if rotating traveling waves are coordinated globally to the same extent in other species, including humans, as they are in mice.

As to the function of rotating wave dynamics, the scientists surmise that they might be serving as space-and-time clocks to set the chain of events of sensation followed by action. The waves could also help pave connections that become more entrenched with practicing a visual-motor task. By streaming across several brain areas, such

waves might provide a way for the brain to begin to predict sensory sequences and coordinate motor responses.

The first author of the paper, Zhiwen Ye, will next set up his own research lab as a junior principal investigator in the Institute of Neuromodulation and Cognition, part of the Shenzhen Medical Academy of Research and Translation, a newly established biomedical research institute in China.

Funding: The research was supported by a National Science Foundation CAREER award (2142911), with additional support from the Pew Biomedical Scholars Program, Klingenstein-Simons Fellowship Award in Neuroscience, National Institutes of Health BRAIN Initiative (U19MH114830), a postdoctoral fellowship from the Washington Research Foundation, and postdoctoral support from National Eye Institute (EY07031).

Key Questions Answered:

Q: How do these newly discovered spiral brain waves physically move through the cortex?

A: The waves travel by following a highly specific, circular architectural layout of neurons within the brain's somatosensory region. The axons of these neurons are physically arranged in a continuous round pattern, very much like a merry-go-round or rail cars sitting on a circular track. This fixed structural pathway naturally

guides the electrical propagation of neural signals into a distinct, rotating vortex pattern that sweeps across space and time.

Q: What major role do these rotating waves play in linking different parts of the brain together?

A: These waves act as a master communications bridge. While they most frequently start in the sensory cortex (the region processing touch, body position, and muscle feedback), they rapidly stream across functional boundaries into the motor cortex. They also mirror perfectly across both hemispheres and align their timing with neural activity deep down in subcortical hubs like the thalamus and striatum, allowing separate but interdependent brain systems to seamlessly share information.

Q: Why do scientists believe these vortex-like waves behave like a “space-and-time clock”?

A: Researchers observed that the waves change characteristics based on an animal’s internal arousal state and its success during coordination tasks. Because they stream fluidly across multiple brain regions, these waves are thought to serve as a spatiotemporal clock that sets the precise sequence of events from sensation to physical action. This constant streaming helps the brain predict upcoming sensory inputs and time its motor responses perfectly.

Editorial Notes:

- Journal paper reviewed in full.
- Additional context added by our staff.

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“[Brainwide topographic coordination of rotating waves](#)” by

Ye Z, Ladd AE, MacKenzie N, Kolich L, Li AJ, Birman D, Bull MS, Daigle TL, Tasic B, Zeng H, Steinmetz NA. *Science*

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Abstract

Brainwide topographic coordination of rotating waves

INTRODUCTION

Electrical activity in the brain often travels in waves, propagating across networks of neurons in patterns that have been linked to sensory perception, memory, and movement. However, the spatial organization of these waves across the brain, the anatomical circuits that give rise to them, and their brain-wide distribution have remained unclear. Understanding these properties is essential for determining the roles that traveling waves may play in behavior and cognition.

RATIONALE

We combined fast, large-scale imaging of neural activity across the mouse cortical surface with high-density electrode recordings in deeper brain structures. This allowed us to track the propagation of traveling waves across the entire cortex while simultaneously measuring the spiking activity of neurons in subcortical regions including the thalamus, striatum, and midbrain.

We also examined the axonal architecture of cortical neurons using three-dimensional reconstructions of their axonal projections, tested the causal role of these circuits within the somatosensory cortex, and measured wave occurrence in different brain states and behavioral contexts.

RESULTS

We discovered that rotating waves, which propagate along a circular trajectory, were a prominent and frequently occurring feature of cortical activity, predominantly centered on the somatosensory cortex. These waves therefore swept sequentially across the maps of the mouse body surface.

The local wiring of neurons in this region displayed a matching circular arrangement, and a computational model confirmed that this architecture supports rotating wave formation. Across the cortex, rotating waves were mirrored between the left and right hemispheres and between sensory and motor areas, reflecting the pattern of long-range connections between these regions.

Severing local circuits within the somatosensory cortex reduced rotating waves in the motor cortex, establishing a mechanistic basis. Subcortical neurons in the thalamus, striatum, and midbrain tracked cortical rotating waves on a moment-to-moment basis in their spiking patterns. Rotating waves were modulated by arousal, evoked by sensory stimulation, and selectively recruited during correct performance of a visual-motor task.

CONCLUSION

These findings reveal that brain activity is shaped by the physical architecture of neural wiring into coordinated rotating waves that span cortical and subcortical regions. Rather than being confined to

isolated brain areas, these waves represent a distributed organizational principle in which the direction and timing of activity propagation are dictated by the geometry of axonal connections.

The recruitment of rotating waves during different behavioral contexts suggests that they may serve as a mechanism for coordinating information flow across sensory and motor systems during perception and action.

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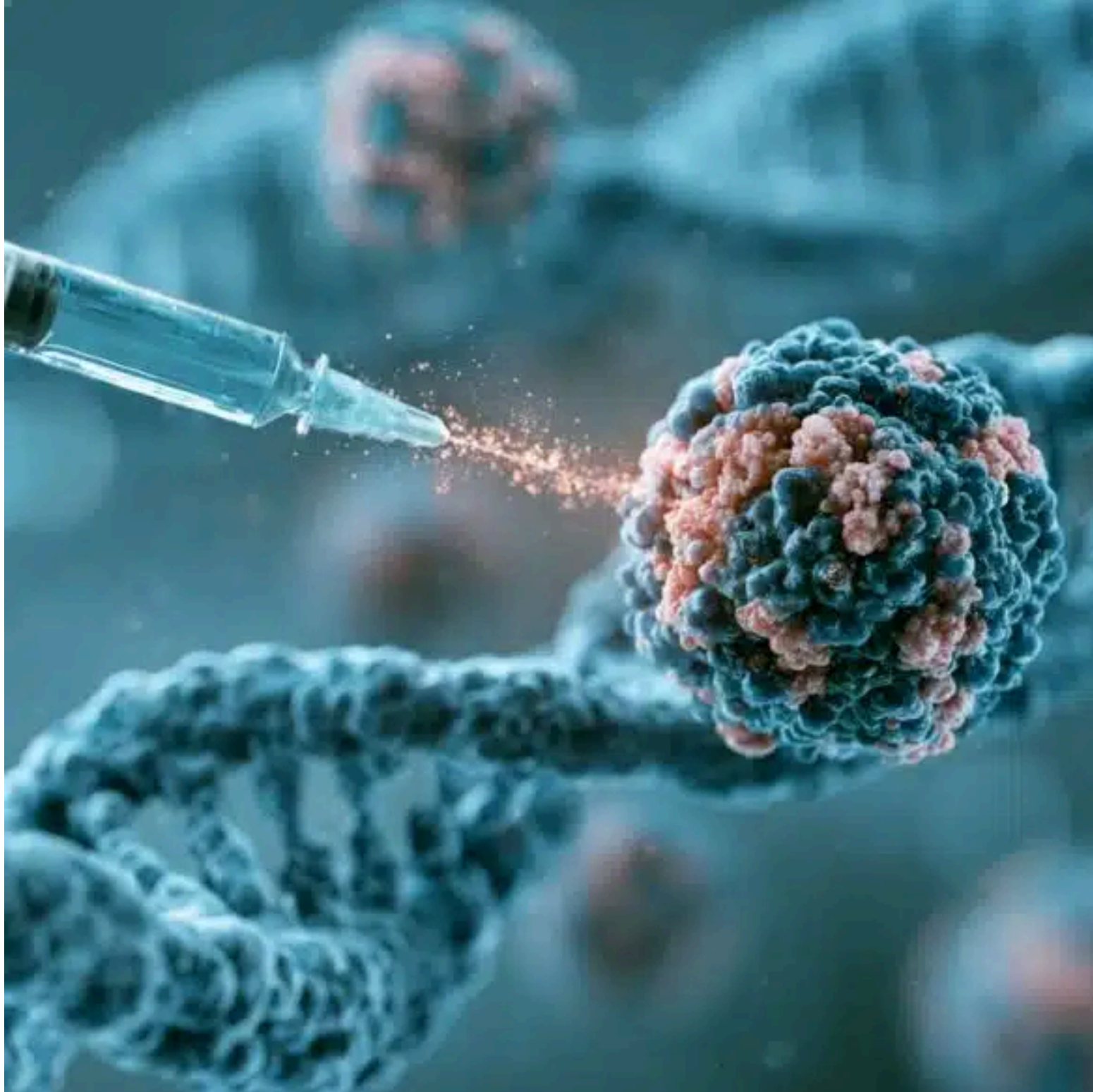
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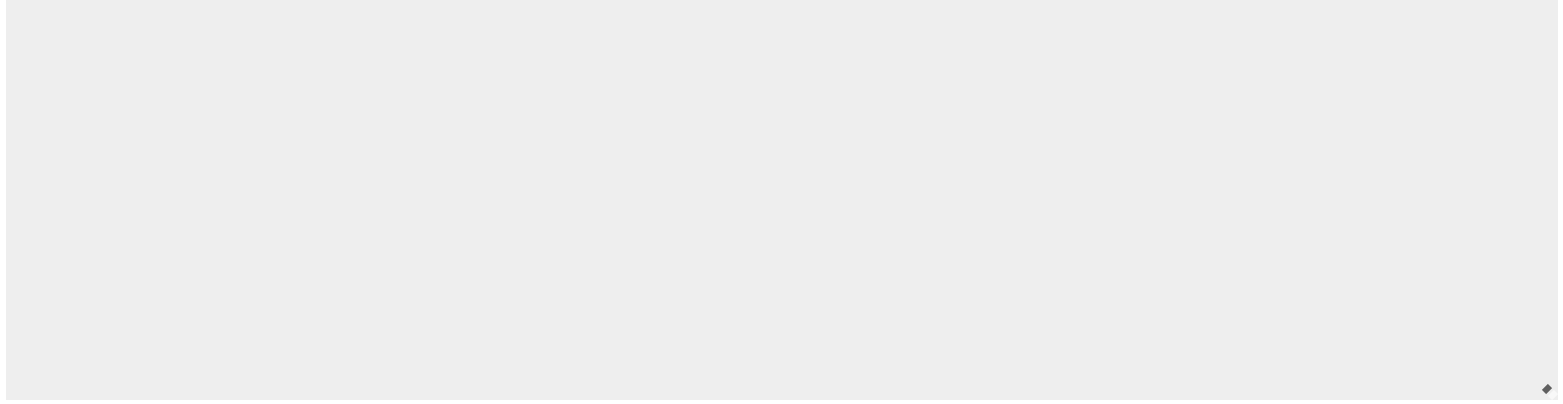


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